

# 1D Helmholtz EM Solver

Design, Validation, and Applications

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APC 523 Final Project

# Problem Overview

**Goal:** solve 1D Helmholtz equation for monochromatic EM scattering at a dielectric transition layer.

## Domain

| Zone       | Range                 | Medium                  |
|------------|-----------------------|-------------------------|
| Region 1   | $z \leq z_1$          | uniform $\varepsilon_1$ |
| Transition | $z_1 \leq z \leq z_2$ | $\varepsilon_r(z)$      |
| Region 2   | $z \geq z_2$          | uniform $\varepsilon_2$ |

**Sign convention:**  $e^{-j\omega t}$ ; right-going  $\propto e^{-jk_z z}$   
 $k_x = k_0 \sqrt{\varepsilon_1} \sin \theta$  (conserved)

## TE (S):

$$\frac{d^2 E_y}{dz^2} + [k_0^2 \varepsilon_r(z) - k_x^2] E_y = 0$$

## TM (P):

$$\frac{d^2 H_y}{dz^2} - \frac{1}{\varepsilon_r} \frac{d\varepsilon_r}{dz} \frac{dH_y}{dz} + [k_0^2 \varepsilon_r(z) - k_x^2] H_y = 0$$

# Two Independent Numerical Methods

| Method       | Back-end              | Order        |
|--------------|-----------------------|--------------|
| ODE shooting | diffraction / JAX GPU | 5th (Dopri5) |
| FEM (P1/P2)  | JAX dense solver      | 2nd / 4th    |

## ODE solvers available:

| Solver            | Family             | Order |
|-------------------|--------------------|-------|
| Heun              | Explicit RK        | 2     |
| Tsit5             | Explicit RK (FSAL) | 5     |
| Dopri5            | Explicit RK (FSAL) | 5     |
| KenCarp4          | IMEX ARK           | 4     |
| SemiImplicitEuler | Symplectic         | 1     |

Butcher Tableaux (values from diffraction)

Heun (Explicit RK2, order 2)

|      | k1     | k2     |
|------|--------|--------|
| c15  | 0      | 0      |
| a100 | 0      | 0      |
| a100 | 0.5000 | 0      |
| b1   | 0.5000 | 0.5000 |
| b2   | 0.5000 | 0      |

Tsit5 (Explicit RK5, FSAL, order 5)

|       | k1     | k2     | k3     | k4     | k5     | k6     | k7     |
|-------|--------|--------|--------|--------|--------|--------|--------|
| c115  | 0      | 0      | 0      | 0      | 0      | 0      | 0      |
| a1000 | 0      | 0      | 0      | 0      | 0      | 0      | 0      |
| a1000 | 0.1667 | 0      | 0      | 0      | 0      | 0      | 0      |
| a1000 | 0.3333 | 0.1667 | 0      | 0      | 0      | 0      | 0      |
| a1000 | 0.5000 | 0.3333 | 0.1667 | 0      | 0      | 0      | 0      |
| a1000 | 0.6667 | 0.5000 | 0.3333 | 0.1667 | 0      | 0      | 0      |
| a1000 | 0.8333 | 0.6667 | 0.5000 | 0.3333 | 0.1667 | 0      | 0      |
| a1000 | 1.0000 | 0.8333 | 0.6667 | 0.5000 | 0.3333 | 0.1667 | 0      |
| b1    | 0.0556 | 0.0556 | 0.0556 | 0.0556 | 0.0556 | 0.0556 | 0.0556 |
| b2    | 0.0556 | 0.0556 | 0.0556 | 0.0556 | 0.0556 | 0.0556 | 0.0556 |

Dopri5 (Dormand-Prince RK5(4), FSAL, order 5)

|       | k1     | k2     | k3     | k4     | k5     | k6     | k7     |
|-------|--------|--------|--------|--------|--------|--------|--------|
| c115  | 0      | 0      | 0      | 0      | 0      | 0      | 0      |
| a1000 | 0      | 0      | 0      | 0      | 0      | 0      | 0      |
| a1000 | 0.0556 | 0      | 0      | 0      | 0      | 0      | 0      |
| a1000 | 0.1111 | 0.0556 | 0      | 0      | 0      | 0      | 0      |
| a1000 | 0.1667 | 0.1111 | 0.0556 | 0      | 0      | 0      | 0      |
| a1000 | 0.2222 | 0.1667 | 0.1111 | 0.0556 | 0      | 0      | 0      |
| a1000 | 0.2778 | 0.2222 | 0.1667 | 0.1111 | 0.0556 | 0      | 0      |
| a1000 | 0.3333 | 0.2778 | 0.2222 | 0.1667 | 0.1111 | 0.0556 | 0      |
| a1000 | 0.3889 | 0.3333 | 0.2778 | 0.2222 | 0.1667 | 0.1111 | 0.0556 |
| a1000 | 0.4444 | 0.3889 | 0.3333 | 0.2778 | 0.2222 | 0.1667 | 0.1111 |
| a1000 | 0.5000 | 0.4444 | 0.3889 | 0.3333 | 0.2778 | 0.2222 | 0.1667 |
| a1000 | 0.5556 | 0.5000 | 0.4444 | 0.3889 | 0.3333 | 0.2778 | 0.2222 |
| a1000 | 0.6111 | 0.5556 | 0.5000 | 0.4444 | 0.3889 | 0.3333 | 0.2778 |
| a1000 | 0.6667 | 0.6111 | 0.5556 | 0.5000 | 0.4444 | 0.3889 | 0.3333 |
| a1000 | 0.7222 | 0.6667 | 0.6111 | 0.5556 | 0.5000 | 0.4444 | 0.3889 |
| a1000 | 0.7778 | 0.7222 | 0.6667 | 0.6111 | 0.5556 | 0.5000 | 0.4444 |
| a1000 | 0.8333 | 0.7778 | 0.7222 | 0.6667 | 0.6111 | 0.5556 | 0.4444 |
| a1000 | 0.8889 | 0.8333 | 0.7778 | 0.7222 | 0.6667 | 0.6111 | 0.5556 |
| a1000 | 0.9444 | 0.8889 | 0.8333 | 0.7778 | 0.7222 | 0.6667 | 0.6111 |
| a1000 | 1.0000 | 0.9444 | 0.8889 | 0.8333 | 0.7778 | 0.7222 | 0.6667 |
| b1    | 0.0556 | 0.0556 | 0.0556 | 0.0556 | 0.0556 | 0.0556 | 0.0556 |
| b2    | 0.0556 | 0.0556 | 0.0556 | 0.0556 | 0.0556 | 0.0556 | 0.0556 |

KenCarp4 — Explicit part (order 4, y=0.25)

|       | k1     | k2     | k3     | k4     | k5     | k6     |
|-------|--------|--------|--------|--------|--------|--------|
| c115  | 0      | 0      | 0      | 0      | 0      | 0      |
| a1000 | 0      | 0      | 0      | 0      | 0      | 0      |
| a1000 | 0.0625 | 0      | 0      | 0      | 0      | 0      |
| a1000 | 0.1250 | 0.0625 | 0      | 0      | 0      | 0      |
| a1000 | 0.1875 | 0.1250 | 0.0625 | 0      | 0      | 0      |
| a1000 | 0.2500 | 0.1875 | 0.1250 | 0.0625 | 0      | 0      |
| a1000 | 0.3125 | 0.2500 | 0.1875 | 0.1250 | 0.0625 | 0      |
| a1000 | 0.3750 | 0.3125 | 0.2500 | 0.1875 | 0.1250 | 0.0625 |
| a1000 | 0.4375 | 0.3750 | 0.3125 | 0.2500 | 0.1875 | 0.1250 |
| a1000 | 0.5000 | 0.4375 | 0.3750 | 0.3125 | 0.2500 | 0.1875 |
| a1000 | 0.5625 | 0.5000 | 0.4375 | 0.3750 | 0.3125 | 0.2500 |
| a1000 | 0.6250 | 0.5625 | 0.5000 | 0.4375 | 0.3750 | 0.3125 |
| a1000 | 0.6875 | 0.6250 | 0.5625 | 0.5000 | 0.4375 | 0.3750 |
| a1000 | 0.7500 | 0.6875 | 0.6250 | 0.5625 | 0.5000 | 0.4375 |
| a1000 | 0.8125 | 0.7500 | 0.6875 | 0.6250 | 0.5625 | 0.5000 |
| a1000 | 0.8750 | 0.8125 | 0.7500 | 0.6875 | 0.6250 | 0.5625 |
| a1000 | 0.9375 | 0.8750 | 0.8125 | 0.7500 | 0.6875 | 0.6250 |
| a1000 | 1.0000 | 0.9375 | 0.8750 | 0.8125 | 0.7500 | 0.6875 |
| b1    | 0.0625 | 0.0625 | 0.0625 | 0.0625 | 0.0625 | 0.0625 |
| b2    | 0.0625 | 0.0625 | 0.0625 | 0.0625 | 0.0625 | 0.0625 |

KenCarp4 — Implicit ESDIRK part (y=0.25 on diagonal)

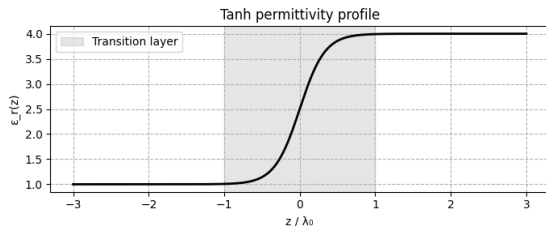
SemiImplicitEuler (Symplectic Euler, order 1)  
pos: forward Euler using vel\_old  
vel: backward-like using pos\_new

# Basic Usage

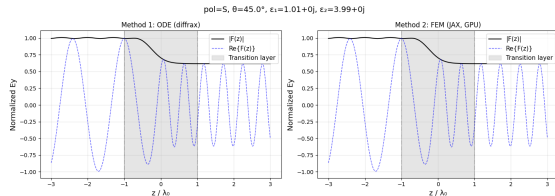
```
from main import HelmholtzSolver1D
import numpy as np

def eps_tanh(z, eps1=1.0, eps2=4.0, zc=0.0, delta=0.33):
    return (eps1+eps2)/2 + (eps2-eps1)/2 * np.tanh((z-zc)/delta)

solver = HelmholtzSolver1D(eps_tanh, z1=-1.0, z2=+1.0,
                           theta_deg=45.0, pol='S', lambda0=1.0)
result_ode = solver.solve_ode()
result_fem = solver.solve_fem(res_trans=200, res_ext=100, order=2)
solver.energy_check(result_ode['r'], result_ode['t']) # R+T should be ~1
solver.plot_field(result_ode)
```



Tanh  $\epsilon_r(z)$ ,  $\epsilon_1=1$ ,  $\epsilon_2=4$



Field profile,  $\theta=45$ , TE

# Correctness: Airy Analytical Benchmark

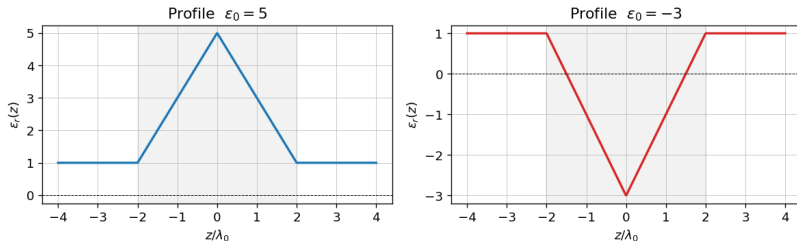
**V-shaped profile** (piecewise-linear):

$$\varepsilon_r(z) = \varepsilon_0 + (1 - \varepsilon_0) |z/\delta|, \quad |z| \leq \delta; \quad \varepsilon_r = 1, \quad |z| > \delta$$

TE equation  $\rightarrow$  Airy equation  $\Rightarrow$  exact closed-form solution in  $\text{Ai}(\xi)$ ,  $\text{Bi}(\xi)$ .

Two test cases ( $\delta = 2\lambda_0$ ):

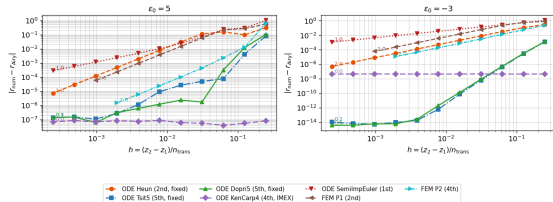
- $\varepsilon_0 = 5$ : dielectric, no turning points
- $\varepsilon_0 = -3$ : metallic core, turning points at  $|z| = 3\delta/4$



V-shaped  $\varepsilon_r(z)$  profiles for both test cases.

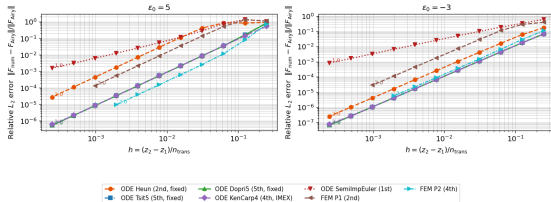
# Correctness: Convergence Results ( $\theta = 45^\circ$ , TE)

Convergence:  $|r_{\text{num}} - r_{\text{Airy}}|$  vs  $h$  — V-profile,  $\theta = 45^\circ$ , TE



$|r_{\text{num}} - r_{\text{Airy}}|$  vs step size  $h$ .

Convergence: field  $L_2$  error vs  $h$  — V-profile,  $\theta = 45^\circ$ , TE



Field  $L_2$  error vs  $h$ .

- **$r$  error:** SemimpEuler slope 1; Heun/FEM P1/P2 slope 2; Dopri5/Tsit5 slope 4 then flattens; KenCarp4 flat (IMEX splitting error).
- **$L_2$  error:** all methods except SemimpEuler converge at slope 2 (limited by 2nd-order quadrature of the norm).
- **Absolute accuracy:** FEM P2 is  $\sim 10^2 \times$  more accurate than P1 at the same  $h$ ; ODE methods match FEM P2 at far coarser resolution. FEM P2 at fine mesh demands tens of GB — ODE is preferred for high accuracy.
- At default ODE resolution ( $50\sqrt{5} \approx 112$  cells/ $\lambda_0$ ):  $|r_{\text{num}} - r_{\text{Airy}}| < 10^{-5}$ .

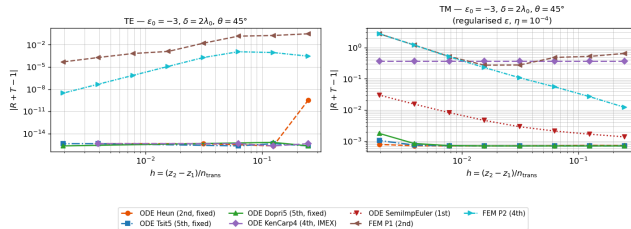
# Stability Near a Dielectric Zero-Crossing

**Setup:**  $\varepsilon_0 = -3$ ,  $\delta = 2\lambda_0$ ,  $\theta = 45^\circ$ , resolution swept 4–512 cells/ $\lambda_0$ .

**TE:** classical turning point ( $R = 1$  exactly from Airy).

**TM:**  $(1/\varepsilon_r)(d\varepsilon_r/dz)$  diverges at  $\varepsilon_r = 0 \Rightarrow$  regularise  $\varepsilon_r \rightarrow \varepsilon_r + j\eta$ ,  $\eta = 10^{-4}$ .

Study 1 — Energy residual  $|R + T - 1|$  vs resolution



Energy residual  $|R + T - 1|$  vs resolution. Left: TE. Right: TM with  $j\eta$  regularisation.

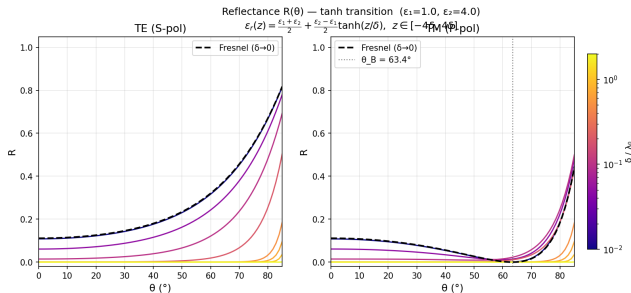
- **TE:** all methods reach floating-point energy conservation; ODE  $|R + T - 1| \lesssim 10^{-14}$ ; FEM improves with  $h$ , P2 > P1.
- **TM:** fixed-step ODE stalls at  $\sim 10^{-3}$ ; KenCarp4 flat at  $\sim 0.5$ .
- All TM methods *worsen* at finer  $h$ ; FEM P2 gives  $|R + T - 1| > 1$  for  $h < 10^{-2}\lambda_0$  —  $j\eta$  regularisation insufficient; reformulation ( $u = H_y/\varepsilon_r$ ) needed.

# Efficiency: GPU-Parallel Parameter Sweeps

Built on **JAX** (64-bit complex); GPU via `jax_platform_name='gpu'`.

`sweep_ode/sweep_fem`: vectorise over  $\theta$  array with `jax.vmap`  $\Rightarrow$  single GPU kernel. **10–50 $\times$  speedup** vs Python loop.

**Example:**  $R(\theta)$  for tanh transition  $\varepsilon_r(z) = \frac{\varepsilon_1 + \varepsilon_2}{2} + \frac{\varepsilon_2 - \varepsilon_1}{2} \tanh(z/\delta)$ ,  $\varepsilon_1=1$ ,  $\varepsilon_2=4$ ,  $\delta$  swept over 8 values  $0.01\lambda_0$ – $2\lambda_0$ .



Reflectance  $R(\theta)$  for TE (left) and TM (right), coloured by  $\delta$ . Black dashed: sharp-interface Fresnel limit.

- TE: reflectance suppressed monotonically as  $\delta$  increases.
- TM: Brewster angle  $\theta_B \approx 63.4$  ( $\varepsilon_2/\varepsilon_1 = 4$ ) preserved for all  $\delta$ .

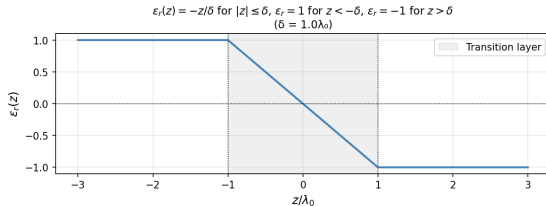


# Application: Plasma Density Diagnostics

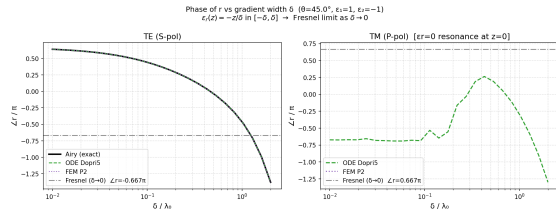
**Profile** (linear ramp into plasma):

$$\varepsilon_r(z) = \begin{cases} 1 & z < -\delta \\ -z/\delta & |z| \leq \delta \\ -1 & z > \delta \end{cases}$$

$\varepsilon_r$  crosses zero at  $z = 0$ . Because  $\varepsilon_2 < 0$ ,  $|r|^2 = 1$  and **phase**  $\angle r$  carries all information about the density gradient  $\delta$ .



$\varepsilon_r(z)$  for  $\delta = \lambda_0$ .



$\angle r$  vs  $\delta$ ,  $\theta = 45$ .

- ODE (TE) agrees with Airy across full  $\delta$  range.
- FEM result not yet correct for this profile (open issue).
- **Prospect:** infer  $\delta$  from measured  $\angle r$ :  

$$\delta_{\text{infer}} = \arg \min_{\delta} |\angle r_{\text{meas}} - \angle r_{\text{ODE}}|^2$$
 applicable to laser-plasma surface diagnostics.

# Known Bugs and Fixes

## Bug 1: Exponential growth for evanescent transmitted wave ( $\varepsilon_2 < 0$ )

*Root cause:* branch cut  $\text{Im}(k_z) \geq 0$  gives  $k_{z2} = +j\alpha$  for plasma. Naïve ansatz  $F \propto e^{-jk_{z2}(z-z_2)} = e^{+\alpha(z-z_2)} \rightarrow \infty$ .

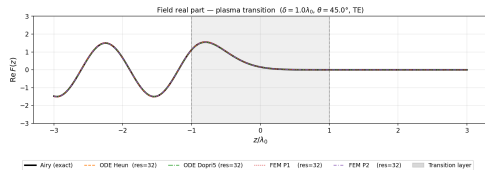
*Fix:* replace  $k_{z2} \rightarrow \overline{k_{z2}}$  everywhere it appears as an outgoing BC (ODE IC, FEM Robin BC, field reconstruction, Airy BC). Backward-compatible:  $k_{z2} \in \mathbb{R}$  for propagating waves.

*Subtlety:* energy check  $|r|^2 = 1$  does *not* catch this bug — only the field profile and  $\angle r$  are corrupted.

## Bug 2: TM instability at $\varepsilon_r = 0$

*Root cause:* drift term  $(1/\varepsilon_r)(d\varepsilon_r/dz)$  diverges.

*Fix:*  $\varepsilon_r \rightarrow \varepsilon_r + j\eta$  ( $\eta = 10^{-4}$ ) shifts pole off real axis. Energy residual converges to floor  $\propto \eta$ ; no blow-up.



$\text{Re } F(z)$  after fix: field decays for  $z > \delta$ .

# Current Limitations and Future Work

## Validation asymmetry

- TE: fully characterised against Airy exact solution.
- TM: verified only by internal consistency (energy check, ODE–FEM agreement); no closed-form reference exists for general  $\varepsilon_r(z)$ .

## Open questions for TM near $\varepsilon_r = 0$

- ① Optimal  $\eta$ : trade-off between numerical stability and physical phase distortion.
- ② Reformulation  $u = H_y/\varepsilon_r$  removes singularity analytically.
- ③ Turning-point asymptotics for the irregular singular point.

## Planned extensions

| Feature                             | Status      | Note                                                     |
|-------------------------------------|-------------|----------------------------------------------------------|
| Gradient-based inversion (autodiff) | Planned     | FEM Jacobian via <code>jax.grad(jnp.linalg.solve)</code> |
| TM analytical reference             | In progress | Riemann–Green function                                   |
| FEM for plasma profile              | Open issue  | $\varepsilon_r = 0$ crossing destabilises assembly       |
| 2D / anisotropic media              | Future      | Coupled TE–TM                                            |